Chapter 16 — Case Studies: Booking & Inventory (Part 2) — Study Edition

> **Where this fits.** Chapter 15 covered the first group of booking problems. This chapter finishes the family with four more: car rental, restaurant, airline and course registration. They are one problem underneath: **a limited resource, a calendar, and many people trying to claim it at once.** Spot that skeleton and every new booking problem is quick to set up.

The shared skeleton of every booking problem

Idea	Car rental	Restaurant	Airline	Course registration
The thing being booked	a vehicle	a table	a seat on one flight instance	a seat in one section
The "calendar"	days	hourly time slots	the flight instance (route + date)	the term
The atomic check-then-claim	`Vehicle.tryBook`	`Table.tryHold`	`FlightInstance.holdSeats`	`Section.enroll`
When it is full	search another car	try the next table	another seat or flight	waitlist
Money	bill lines + late fee	check split by seat + tax	fare per seat class, paid per itinerary	none (fees are an extension)

A **resource** is anything in limited supply. A **race condition** is a bug where the result depends on which thread runs first — here, two people both "win" one seat because both checked "free?" before either wrote "taken".

How to run: `cd chapters/16-case-booking-2/code && javac --release 21 -d /tmp/lld_16 $(find . -name '*.java') && java -cp /tmp/lld_16 lld.ch16.carrental.Main` (or `restaurant`, `airline`, `courseregistration`). Where threads race, the *counts* are always the same; *which* racer wins varies. One file sometimes holds several small package-private types (legal Java; production code would split them).

> **Money.** All prices use `BigDecimal`, never `double` — a `double` cannot store 0.1 exactly, so sums drift by fractions of a cent.

Design a Car Rental System

Problem & clarifying questions

A **car rental system** rents vehicles for hours to weeks from many branches; a car may be returned to a different branch.

Question	Assumed answer
Vehicle kinds?	Car, truck, SUV, van, motorcycle.
Branches?	Many, each with its own vehicles; drop-off branch may differ from pick-up.
Price?	Daily rate per vehicle type + add-ons + a late fee per day late.
Add-ons?	Insurance, equipment (navigation, child seat, ski rack), services (roadside assistance, wifi), additional drivers.
Identification?	Unique barcode, scanned at pick-up and return.
History?	A vehicle log of every event (reserved, picked up, returned, fuelling, cleaning, damage).
Concurrency?	Two members must never get the same car on overlapping days.

Requirements

Functional: (1) rent car/truck/SUV/van/motorcycle; (2) add each vehicle with a unique barcode and details incl. parking stall; (3) answer "who took this vehicle?" and "what has this member rented?"; (4) late fee after the due date; (5) search and reserve any available vehicle; (6) notify when pick-up is near, due date is near, and when overdue; (7) read barcodes; (8) cancel reservations; (9) vehicle log; (10) rental insurance; (11) extra equipment; (12) services.

Non-functional: no double booking for overlapping days; new add-ons or pricing rules without changing reservation code (Open/Closed); auditability — every charge is a bill line, every event a log entry.

Use cases

Actors: **Receptionist** (adds/edits vehicles and workers, can reserve for a customer), **Member** (searches, reserves, picks up, returns), **Worker** (handles returns, updates the vehicle log), **System** (reminders, overdue alerts, cancellations). Top use cases: add/edit vehicle; search by type and availability; register/cancel membership; reserve; check out; return; add equipment/services; update log. Reserving is shared by member and receptionist; the log belongs to the worker.

Core entities

Class	Responsibility
`CarRentalSystem`	Owns the branches and the reservation service.
`CarRentalLocation`	One branch with its own `VehicleInventory`.
`VehicleInventory`	Indexes by barcode, type and model; "which SUVs are free on these days?".
`Vehicle`	Barcode, plate, type, parking stall, mileage, **the set of booked days**, its log.
`VehicleLogEntry`	One event (date, type, note).
`DateRange`	Half-open `[start, end)`; `end` is the due date.
`Reservation`	Member, vehicle, dates, pick-up/return branch, add-ons, drivers, bill, status; holds the business rules.
`ReservationAddOn` (+ `RentalInsurance`, `Equipment`, `Service`), `AdditionalDriver`	Things that add a charge; a named extra driver (no charge).
`Bill`, `BillItem`	The one place that knows what a rental costs.
`PricingStrategy`, `LateFeePolicy`	Pluggable base-charge and late-fee rules.
`ReservationService`	Creates reservations atomically, answers "who rented what", runs reminders.
`Notifier`, `PaymentMethod`	Ports for messages and money.

Class diagram

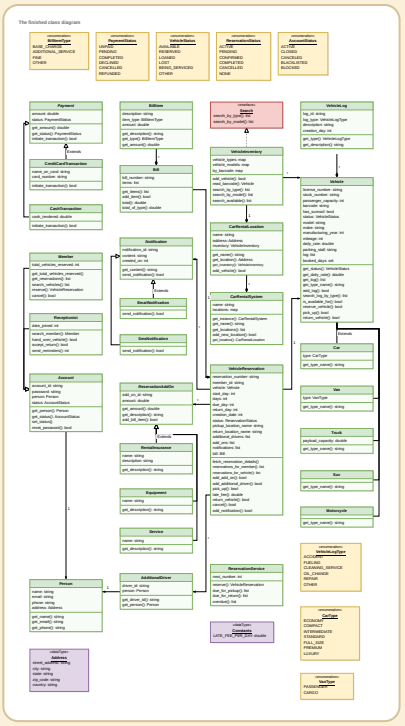
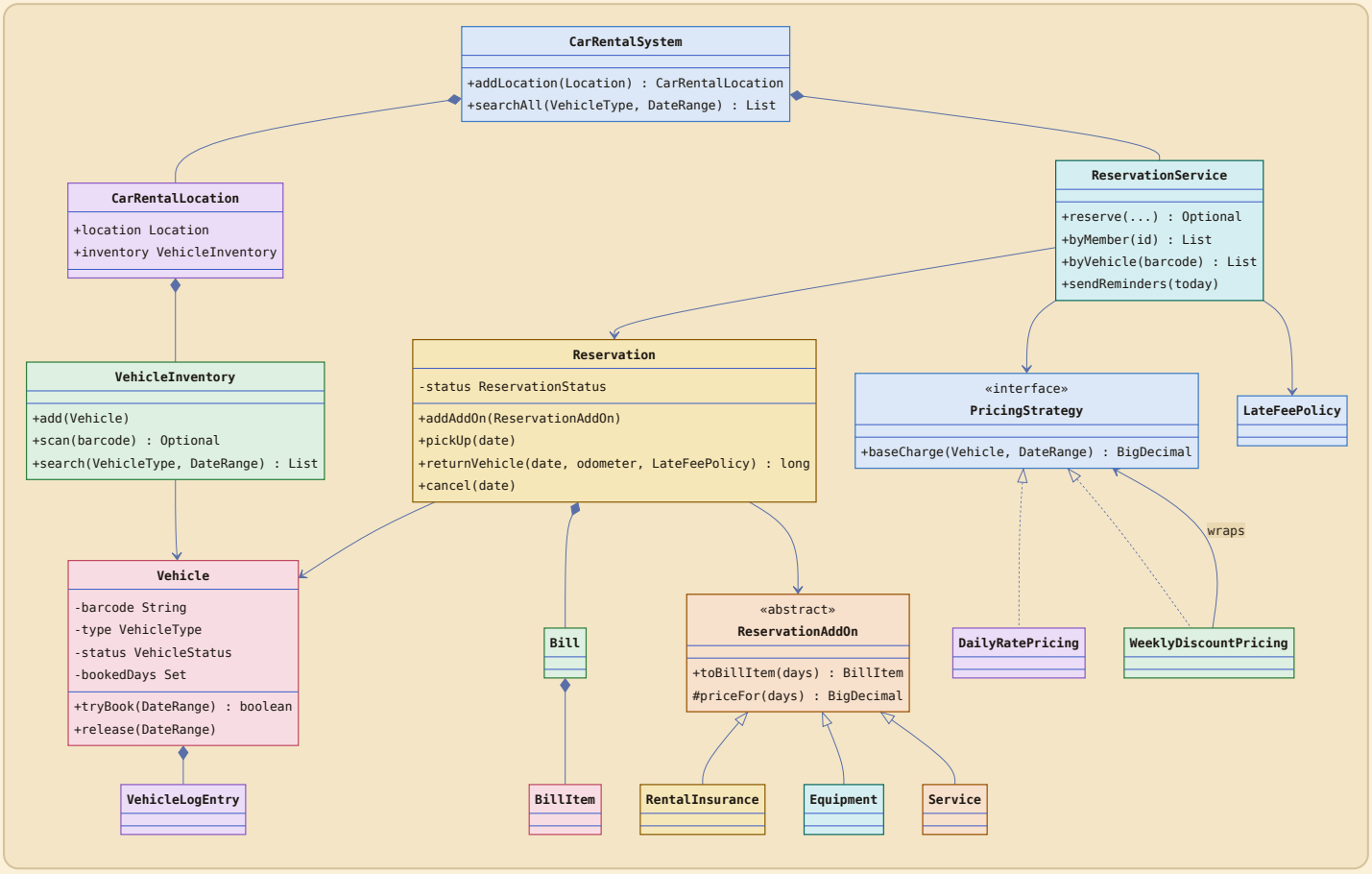


Figure — The finished class diagram of the car rental system (course diagram)

The system owns its branches and each branch owns its inventory (**composition** — the part dies with the whole); the inventory only *refers to* vehicles (**association**). A reservation owns its bill but refers to the vehicle. Our Java design differs in two ways: vehicle kinds are an enum (the five subclasses added no behaviour), and pricing is a strategy object.



Key flows

- **Pick up:** scan licence (or search by name) → check reservation → scan vehicle barcode → hand over keys; no reservation → reserve a free car on the spot.
- **Return:** worker scans the barcode → past due date adds a late fee (one line in `returnVehicle`) → check damage and fuel → settle the bill.
- **Reserve:** `ReservationService.reserve` calls `vehicle.tryBook(dates)`; if false the member picks another car; if true it prices via `PricingStrategy`, creates the reservation (bill gets a BASE line) and notifies.
- **Lifecycle (state machine):** `CONFIRMED →pickUp→ ACTIVE →returnVehicle→ COMPLETED`; `CONFIRMED →cancel→ CANCELLED` (days released on cancel and return).

Design decisions & patterns used

Decision	Why	Alternative rejected
Availability = booked days per vehicle, not a status	Members ask about future dates; a car loaned now may be free next week, a car on the lot may be promised next week.	A `RESERVED` status (can't answer date questions; one reservation per car).
Keep `VehicleStatus` too	Staff need "where is it now" (lot, loaned, in service, lost).	Dropping it.
`tryBook` checks **and** books in one `synchronized` method	Closes the check-then-act race.	`isFree()` then `book()`.
Template Method on `ReservationAddOn`	Billing written once in `final toBillItem`; subclasses give `kind` and may override the `priceFor` hook.	Type enum + `switch`.
Strategy for pricing and late fee; **Decorator** `WeeklyDiscountPricing`	Swap/stack rules without touching the reservation.	Constants; a subclass per combination.
`Bill` is the only place charges live; methods `synchronized`, `settle` idempotent	One total; no line added mid-payment; a double-click pays once.	Per-add-on running totals.
`ReservationService` separate from `Reservation`	Creating/indexing vs one reservation's rules (SRP).	Static factory.
Actors call the rules, never compute them	Member calls the service; receptionist calls the reservation.	`Receptionist.computeLateFee()`.
Course's `CarRentalSystem` **Singleton** → we pass one in	Same "one per app", but testable (dependency injection).	`getInstance()`.

Java implementation

Package `lld.ch16.carrental`. Full runnable code: code/lld/ch16/carrental/ (see the Full Edition).

```java
// Vehicle.java — check AND book inside one lock
synchronized boolean isFreeFor(DateRange r) {
    return r.dates().noneMatch(bookedDays::contains);
}

synchronized boolean tryBook(DateRange r) {
    if (!isFreeFor(r)) return false;
    r.dates().forEach(bookedDays::add);
    return true;
}

synchronized void release(DateRange r) { r.dates().forEach(bookedDays::remove); }

// Billing.java — Template Method
abstract class ReservationAddOn {
    // ...
    final BillItem toBillItem(long rentalDays) {
        return new BillItem(kind() + ": " + name, BillItemType.ADD_ON, priceFor(rentalDays));
    }

    protected abstract String kind();

    /** Hook: by default an add-on is charged per rental day. */
    protected BigDecimal priceFor(long rentalDays) { return price.multiply(BigDecimal.valueOf(rentalDays)); }
}

/** A service such as roadside assistance or wifi is a flat fee per rental. */
final class Service extends ReservationAddOn {
    Service(String name, BigDecimal flatFee) { super(name, flatFee); }
    protected String kind() { return "Service"; }
    @Override protected BigDecimal priceFor(long rentalDays) { return price; }
}

// Pricing.java — Decorator over any pricing: 10% off rentals of 7 days or more.
final class WeeklyDiscountPricing implements PricingStrategy {
    private final PricingStrategy inner;
    WeeklyDiscountPricing(PricingStrategy inner) { this.inner = inner; }
    public BigDecimal baseCharge(Vehicle v, DateRange p) {
        BigDecimal base = inner.baseCharge(v, p);
        return p.days() >= 7 ? base.multiply(new BigDecimal("0.90")).setScale(2, RoundingMode.HALF_UP) : base;
    }
}

// Reservation.java — the rules live here
synchronized long returnVehicle(LocalDate on, int odometer, LateFeePolicy lateFee) {
    require(ReservationStatus.ACTIVE, "return");
    long daysLate = Math.max(0, ChronoUnit.DAYS.between(period.end(), on));
    if (daysLate > 0) {
        bill.add(new BillItem("Late fee: " + daysLate + " day(s)", BillItemType.LATE_FEE,
                lateFee.fee(vehicle, daysLate)));
    }
    vehicle.setMileage(odometer);
    vehicle.setStatus(VehicleStatus.AVAILABLE);
    vehicle.release(period);
    // ... log RETURNED, status = COMPLETED
}

// RentalSystem.java — ReservationService
Optional<Reservation> reserve(Member m, Vehicle v, DateRange period, Location pickup, Location dropoff, LocalDate
today) {
    if (!v.tryBook(period)) return Optional.empty();
    var r = new Reservation("R" + seq.incrementAndGet(), m, v, period, pickup, dropoff, pricing.baseCharge(v,
period));
    reservations.put(r.id(), r);
    v.addLog(today, VehicleLogType.RESERVED, r.id() + " for " + period);
    notifier.send(m, "Reservation " + r.id() + " confirmed: " + v.model() + " " + period);
    return Optional.of(r);
}
```

Output excerpt:

```
3) 8 members race for VH-002 on the same dates
   successful reservations for VH-002: 1
6) Alice's bill
   Base: RAV4 x 3 days                      BASE_CHARGE   240.00
   Insurance: Collision damage waiver       ADD_ON        45.00
   Equipment: Child seat                    ADD_ON        24.00
   Service: Roadside assistance             ADD_ON        25.00
   Late fee: 2 day(s)                       LATE_FEE      80.00
   TOTAL 414.00
   paid: true, settle again (double click) -> true, no second charge
10) A 7-day Civic rental gets the weekly discount
   base charge 315.00 (7 x 50 = 350, minus 10%)
```

Eight threads race for one car and exactly one wins; the late fee is 2 × 40.00; a second `settle` does not charge again.

Concurrency & edge cases

- **Double booking** (a gap the course admits): `tryBook` is `synchronized` per vehicle, so check and mark happen as one step; per-vehicle locks let different cars book in parallel. In a database: unique key on `(vehicle_id, day)` or `SELECT … FOR UPDATE`.
- **Search is advisory** — `reserve` may still fail; show "no longer available".
- **Late return collides with the next rental** — keep a buffer day; reassign the next booking to another car of the same type.
- **Early return** releases the whole range; refunds are policy. **Cancel after pick-up**, **add-ons after the rental**, **invalid ranges**, **odometer going backwards** are all rejected.
- **Different return branch** — recorded, but the car stays in the pick-up inventory: a known gap (transfers).

Extensions & follow-up questions

- **One-way rentals?** On return at another branch move the vehicle between inventories, log a `TRANSFER`, add a one-way fee line.
- **Seasonal/dynamic pricing?** A `SeasonalPricing` strategy or another decorator; reservation unchanged.
- **Deposits?** `authorize` at pick-up, `capture` at return. **Mileage/fuel/damage?** `FINE` bill lines. **Reminders?** A daily scheduler calls `sendReminders(today)`. **Scale?** Partition by branch, `(vehicle_id, day)` unique key, search on a replica/index.

In the interview

30-second answer: "Branches own inventories indexed by barcode, type and model. Availability is a question about *dates*, so each vehicle keeps its booked days and `tryBook` checks and books atomically. A reservation owns a bill; every charge is a line. Add-ons share a template-method base; pricing and late fees are strategies."

Plan (40 min): 0–5 clarify; 5–10 actors and entities; 10–20 class diagram ("dates, not status"); 20–35 code `tryBook`, `Reservation`, `reserve`; 35–40 concurrency, late collisions, transfers. Likely follow-ups: book next month while loaned now? (yes — dates); two clicks at once? (atomic `tryBook`); new add-on? (one subclass).

Self-check

1. **Q:** Why isn't a `RESERVED` status enough for "free next week?" **A:** Status describes *now*; only per-day bookings answer date questions.
2. **Q:** Pattern and hook in `toBillItem`? **A:** Template Method; `priceFor` is the hook (`Service` makes it flat).
3. **Q:** Why one `tryBook` not `isFree` + `book`? **A:** Atomic check-then-act under the vehicle lock.
4. **Q:** How is the late fee computed? **A:** Days from `period.end` to return × `LateFeePolicy` rate, as a `LATE_FEE` line.

TL;DR

- Availability = booked **days** per vehicle; status = where it is now.
- `tryBook` = atomic check-and-book per vehicle.
- One bill per reservation; late fee = days late × rate.
- Template Method (add-ons), Strategy + Decorator (pricing). Gaps: branch transfers, buffer between rentals.

Real-world example. Hertz or Enterprise sell a *car class* ("intermediate SUV or similar"), counting availability per type per day and binding a specific vehicle at the counter — the same design one level up.

Design a Restaurant Management System

Problem & clarifying questions

A **restaurant management system** runs tables, reservations, orders, the kitchen, and the bill.

Question	Assumed answer
One restaurant or chain?	Several branches, each with its own menu, tables, staff.
Menu?	Menu → sections → items with prices.
Orders?	Waiter creates an order per table and adds meals **per seat**; meal items point at menu items.
Reservations / walk-ins?	Search tables by time and party size and reserve; cancel; reminders; show tables free now.
Layouts?	Several seating arrangements per branch ("Weekday", "Banquet").
Payment?	Card, cheque or cash.
Kitchen?	Orders go to a chef queue, cooked in order; waiter told when ready.

Requirements

Functional: (1) several branches; (2) menu per branch; (3) sections with items; (4) waiter creates an order per table, meals per seat; (5) meal items ↔ menu items; (6) free tables for walk-ins; (7) table reservations; (8) search by date/time and reserve; (9) cancel; (10) reminder notification; (11) pay by card, cheque or cash; (12) several seating arrangements; (13, added) kitchen queue, waiter notified when ready.

Non-functional: no two parties on one table for overlapping time; prices live in one place (the menu item); easy to add a payment type or chef-assignment rule.

Use cases

Actors: **Receptionist** (tables, layouts, reservations, check-in), **Waiter** (orders), **Manager** (staff, menu), **Chef** (cooks), **Cashier** (check and payment), **System** (reminders). Top use cases: add/modify tables, search tables, place/update order, create/cancel reservation, check in, pay. Each role has a small separate set — each role class gets only its own methods.

Core entities

Class	Responsibility
`Restaurant`	Owns branches; closing it deactivates every employee account.
`Branch`	Owns menu, kitchen, seating charts, reservation service; employs staff.
`Menu`, `MenuSection`, `MenuItem`	Menu tree; **price only on `MenuItem`**.
`Table`, `TableSeat`	Capacity and seats; **current status** and **held time slots**.
`TableChart`	Named arrangement; refers to tables (a table can be in several charts).
`Reservation`	Customer, party size, time, table, status; `updatePeopleCount` refuses more than the table seats.
`ReservationService`	Best-fit search, atomic reserve, cancel, check-in, reminders, walk-ins.
`Order`, `Meal`, `MealItem`	One order per visit; one meal per seat; item × quantity; status machine + listeners.
`Kitchen`, `Chef`	Kitchen picks a chef by a pluggable rule; each chef has a FIFO queue.
`Employee` + roles	`Receptionist`, `Waiter`, `Manager`, `Chef`, `Cashier`.
`Check`	Subtotal, tax, total — computed from the order, never stored.
`Payment` (`CardPayment`, `CashPayment`, `CheckPayment`)	Three behaviours.

Class diagram

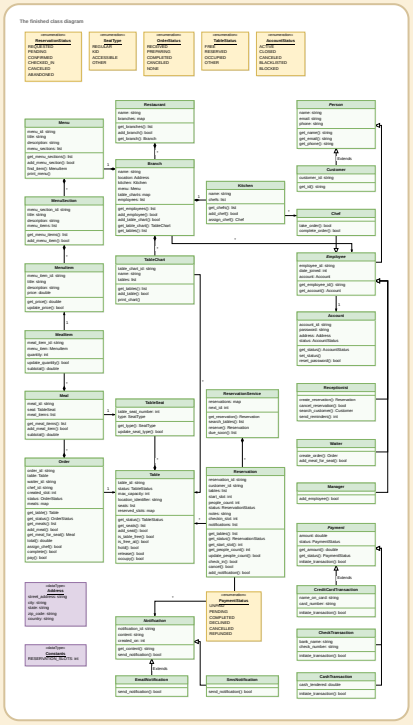
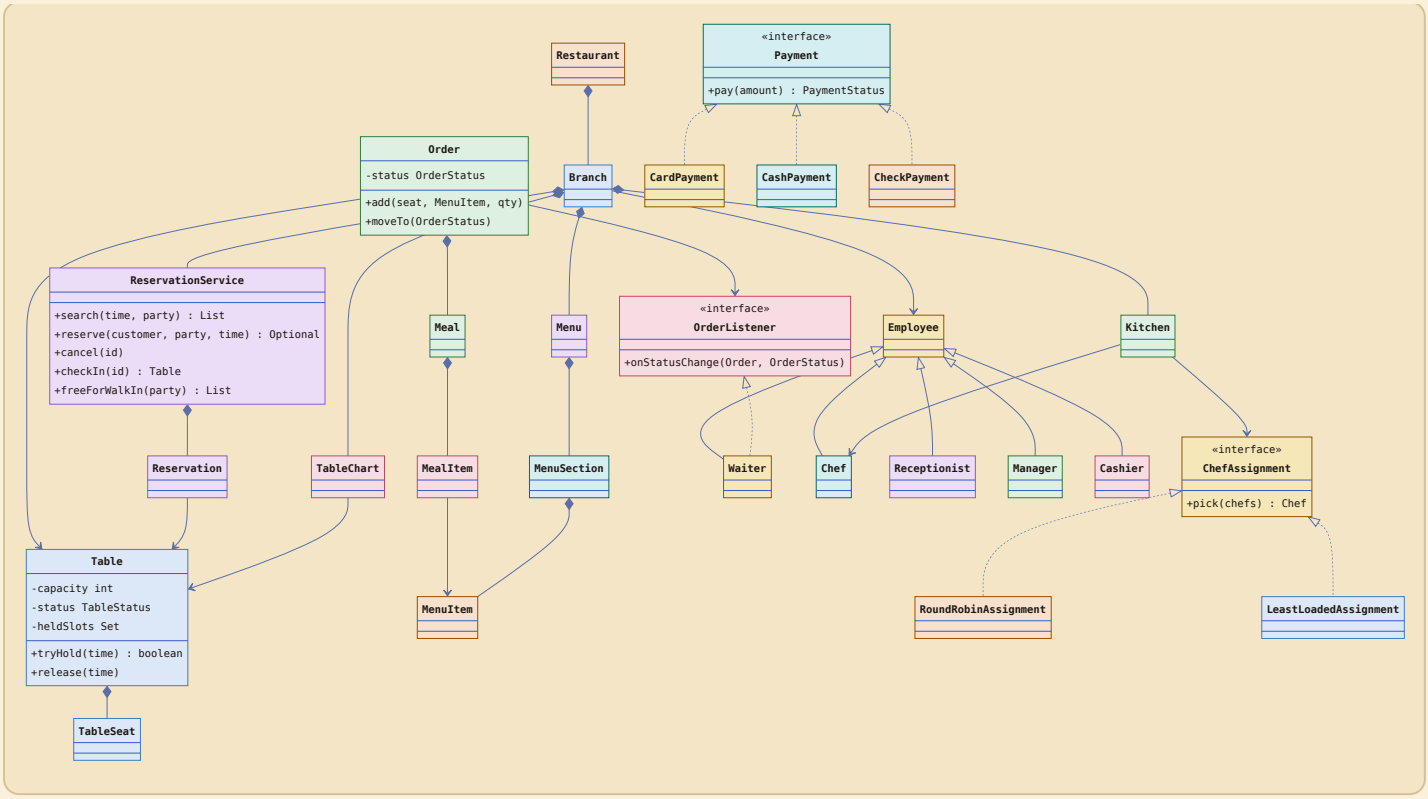


Figure — The finished class diagram of the restaurant management system (course diagram)

The branch is the hub: menu, kitchen and charts are composition; staff are associated (they can move branches). There is **no stored `Bill`** — the order already holds every priced line, so a stored total could disagree. Our `Check` is computed on demand.



Key flows

- **Place an order:** waiter opens an order for the table, adds a meal per seat, submits it to the kitchen.
- **Make a reservation:** search by time and party size → reserve and confirm, or "none available". **Cancel:** look up, cancel, release table, notify.
- **Order lifecycle:** `RECEIVED → PREPARING` (no more items) `→ READY` (waiter paged) `→ SERVED` (cashier settles); `RECEIVED → CANCELLED`.
- **Order to payment:** `Kitchen.submit` → `ChefAssignment.pick` → chef queue → `PREPARING` → `READY` → Observer pages the waiter → `Check.of(order)` → `payment.pay(total)` → `SERVED`, table `FREE`.

Design decisions & patterns used

Decision	Why	Alternative rejected
One meal per seat (`Map<seat, Meal>`)	Plate per guest; split bills are trivial.	Flat list of dishes.
Table keeps held slots as well as status	Status = now (walk-ins); slots = future bookings (a booking holds two hourly slots).	Status only.
Search on `ReservationService`, not `Table`	One table knows only itself (an old course version had an unworkable static `Table.search`).	Static search.
Best fit	Smallest table that fits keeps big tables for big parties.	First free.
Observer (`OrderListener`)	Waiter, screen, delivery app can all subscribe to "ready".	Chef calls waiter directly.
Strategy (`ChefAssignment`)	Course rotates chefs (fair, not load-aware); `LeastLoadedAssignment` is one class.	Hard-coded round-robin.
State machine in `Order.moveTo`	Illegal moves fail loudly.	Free `setStatus`.
`sealed` `Payment` with three classes	Card completes, cash may be short/gives change, cheque is pending.	One class + `if`s.
Check computed; `Customer` not an `Employee`	No second total; only staff have accounts.	Stored bill; one `Person` with a role flag.

Java implementation

Package `lld.ch16.restaurant`. Full runnable code: code/lld/ch16/restaurant/ (see the Full Edition).

```java
// Tables.java — Atomic check-then-hold of two hourly slots
synchronized boolean isFreeAt(LocalDateTime start) { return slots(start).stream().noneMatch(heldSlots::contains);
}

synchronized boolean tryHold(LocalDateTime start) {
    if (!isFreeAt(start)) return false;
    heldSlots.addAll(slots(start));
    return true;
}

// Reservations.java — Best fit first
List<Table> search(LocalDateTime time, int partySize) {
    return tables.stream().filter(t -> t.capacity() >= partySize && t.isFreeAt(time))
            .sorted(Comparator.comparingInt(Table::capacity)).toList();
}

Optional<Reservation> reserve(Customer c, int partySize, LocalDateTime time) {
    for (Table t : search(time, partySize)) {
        if (t.tryHold(time)) {     // another receptionist may have taken it since search(): try the next one
            var r = new Reservation("RES-" + seq.incrementAndGet(), c, partySize, time, t);
            reservations.put(r.id(), r);
            notifier.send(c, "Table " + t.id() + " booked for " + partySize + " at " + time);
            return Optional.of(r);
        }
    }
    return Optional.empty();
}

// Orders.java — state machine + Observer
void moveTo(OrderStatus next) {
    synchronized (this) {
        boolean ok = switch (status) {
            case RECEIVED -> next == OrderStatus.PREPARING || next == OrderStatus.CANCELLED;
            case PREPARING -> next == OrderStatus.READY;
            case READY -> next == OrderStatus.SERVED;
            case SERVED, CANCELLED -> false;
        };
        if (!ok) throw new IllegalStateException("order " + id + ": " + status + " -> " + next + " not allowed");
        status = next;
    }
    listeners.forEach(l -> l.onStatusChange(this, next));   // notify outside the lock
}

// Kitchen.java — Strategy: which chef gets the next order
interface ChefAssignment { Chef pick(List<Chef> chefs); }

final class LeastLoadedAssignment implements ChefAssignment {
    public Chef pick(List<Chef> chefs) { return
chefs.stream().min(Comparator.comparingInt(Chef::load)).orElseThrow(); }
}

// Staff.java — Cashier
PaymentStatus settle(Check check, Payment payment) {
    // Check the order state BEFORE taking money: a refused state change after payment would leave the guest
charged.
    if (check.order().status() != OrderStatus.READY)
        throw new IllegalStateException("order " + check.order().id() + " is " + check.order().status() + ", not
READY");
    var status = payment.pay(check.total());
    if (status != PaymentStatus.DECLINED) {
        check.order().moveTo(OrderStatus.SERVED);
        check.order().table().setStatus(TableStatus.FREE);
    }
    return status;
}
```

Output excerpt:

```
2) Ravi searches tables for 4 at 19:00: [T2, T3, T4]
   Asha -> T2, Ben -> T3 (T2 is held 19:00-21:00)
   Asha's party grows to 6: rejected, table T2 seats 4, not 6
6) Kitchen works the queues
   rejected: order O1 is already PREPARING
    [Wendy's pager] order O1 for table T2 is READY
7) Check for O1 (split by seat)
   seat 1: [1x Tomato soup, 1x Margherita pizza] = 18.50
   subtotal 43.00, tax 3.44, total 46.44
   O2 total 25.92 paid by check: PENDING
```

Concurrency & edge cases

- **Two receptionists, one table:** search is advisory; `tryHold` checks and holds both slots atomically; on failure try the next best fit.
- **Notify outside the lock:** calling unknown listener code while locked risks deadlock and long waits.
- **Reminder spam:** `markReminded` sends once per reservation.
- **Party larger than any table:** "no single table fits" (combining tables is an extension). **No-shows:** `NO_SHOW` status; a job frees tables 15 min after start.
- **Party grows** beyond the table → rejected. **Pay before ready** → `settle` checks `READY` before taking money. Unknown id → clear `IllegalArgumentException`; bad seat number rejected.
- **Cash short** → `DECLINED`, order stays `READY`. **Cheque** → `PENDING`.

Extensions & follow-up questions

- **Large party, two tables?** Reservation holds `List<Table>`; lock tables in a fixed order (by id) to avoid deadlock.
- **Split bill?** Already easy — per-seat `Meal.subtotal()`. **Tips?** Extra `Check` lines. **Walk-in waitlist?** FIFO per party size.
- **Kitchen stock?** Ingredient `Inventory`, recipes, mark items "86'd". **Real kitchen threads?** Each chef takes from a `BlockingQueue<Order>`. **Shifts?** A `Shift` entity; only on-shift chefs are picked.

In the interview

30-second answer: "Branch owns a menu tree, kitchen, charts and reservation service. Tables keep status for walk-ins and held slots for bookings; the service searches best-fit and holds atomically. Orders have a meal per seat, a strict state machine and an observer for 'ready'. Chefs are assigned by a strategy. The check is computed; payment is card, cash or cheque strategies."

Plan: 0–5 clarify; 5–10 six actors; 10–20 diagram ("meal per seat", "status vs slots"); 20–35 code `tryHold`, `reserve`, `Order`, `Kitchen.submit`; 35–40 races, large parties, load balancing.

Self-check

1. **Q:** Why both status and held slots? **A:** Status = now (walk-ins); slots = future times (reservations).
2. **Q:** Why is search on the service? **A:** One table only knows itself.
3. **Q:** Which pattern pages the waiter, and why notify outside the lock? **A:** Observer; foreign code under a lock risks deadlock.
4. **Q:** Why is a cheque `PENDING`? **A:** It must clear at the bank.

TL;DR

- Menu → section → item; price only on the item.
- Table = status (now) + held slots (later); best-fit, atomic hold.
- Order → meal per seat; state machine; Observer for "ready"; Strategy for chefs.
- Check is computed; card/cash/cheque strategies.

Real-world example. POS systems like Toast send orders as tickets to kitchen display screens in order (FIFO + Observer); OpenTable models inventory by time slot and party size, like `heldSlots`.

Design an Airline Management System

Problem & clarifying questions

An **airline management system** schedules flights, reserves tickets, cancels flights, and manages staff and crew.

Question	Assumed answer
Search by?	Date, source and destination airport.
Trips?	An **itinerary** of several flights, several passengers (no accounts needed).
Seat classes/types?	Economy, economy plus, preferred economy, business, first; regular, accessible, emergency exit, extra legroom.
Who manages flights?	Only the admin adds aircraft/flights/schedules and cancels flights; everyone affected is notified.
Crew?	Pilots and crew assigned to flights, can view their schedule.
Payment / notifications?	Paid per itinerary; notify on booking changes and flight status changes.

Requirements

Functional: (1) search by date, source, destination; (2) reserve tickets, build itineraries; (3) view schedules and available seats; (4) several passengers per itinerary; (5) only the admin adds aircraft, flights, schedules and cancels flights (all notified); (6) customers cancel a reservation or itinerary; (7) assign pilots and crew; (8) take payment; (9) notify on booking create/modify and flight status change.

Non-functional: a seat on an instance is sold at most once; a multi-passenger booking gets all seats or none (**atomic**); fare rules change without touching booking code.

Use cases

Actors: **Admin**, **Front desk officer** (books for walk-ins), **Customer**, **Pilot/Crew** (view assignments), **System** (notifications). Top use cases: search; create/modify/view reservation; assign seats; pay; update schedule; assign crew. The officer's booking is the same operation as the customer's, triggered by another actor.

Core entities

The key is **four "seat-ish" and "flight-ish" ideas**:

Idea	Class	Example
The route	`Flight`	GA212 London → Tokyo, 14 h, base fare 500
The route on one date	`FlightInstance`	GA212 on 8 Oct 2026 at 10:00
The physical seat	`Seat`	12A, extra legroom, economy plus
That seat on that instance	`FlightSeat`	12A on GA212 8 Oct — booked by PNR147

Others: `Airport`; `Aircraft` (registration, model, seats); `WeeklySchedule` ("every Tuesday 10:00"); `Passenger` (a name on a ticket — **not** an account); `Customer` (books and pays); `Admin`, `FrontDeskOfficer` (admin-only methods need an `Admin` argument); `CrewMember`; `FlightReservation` (one leg: instance, passenger → seat, amount, status); `Itinerary` (legs, connection rule, the unit paid); `FareStrategy`; `Airline` (Facade, keeps the index instance → reservations).

Class diagram

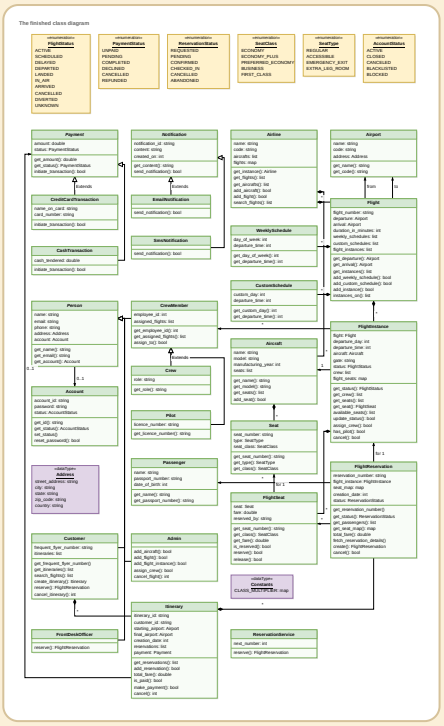
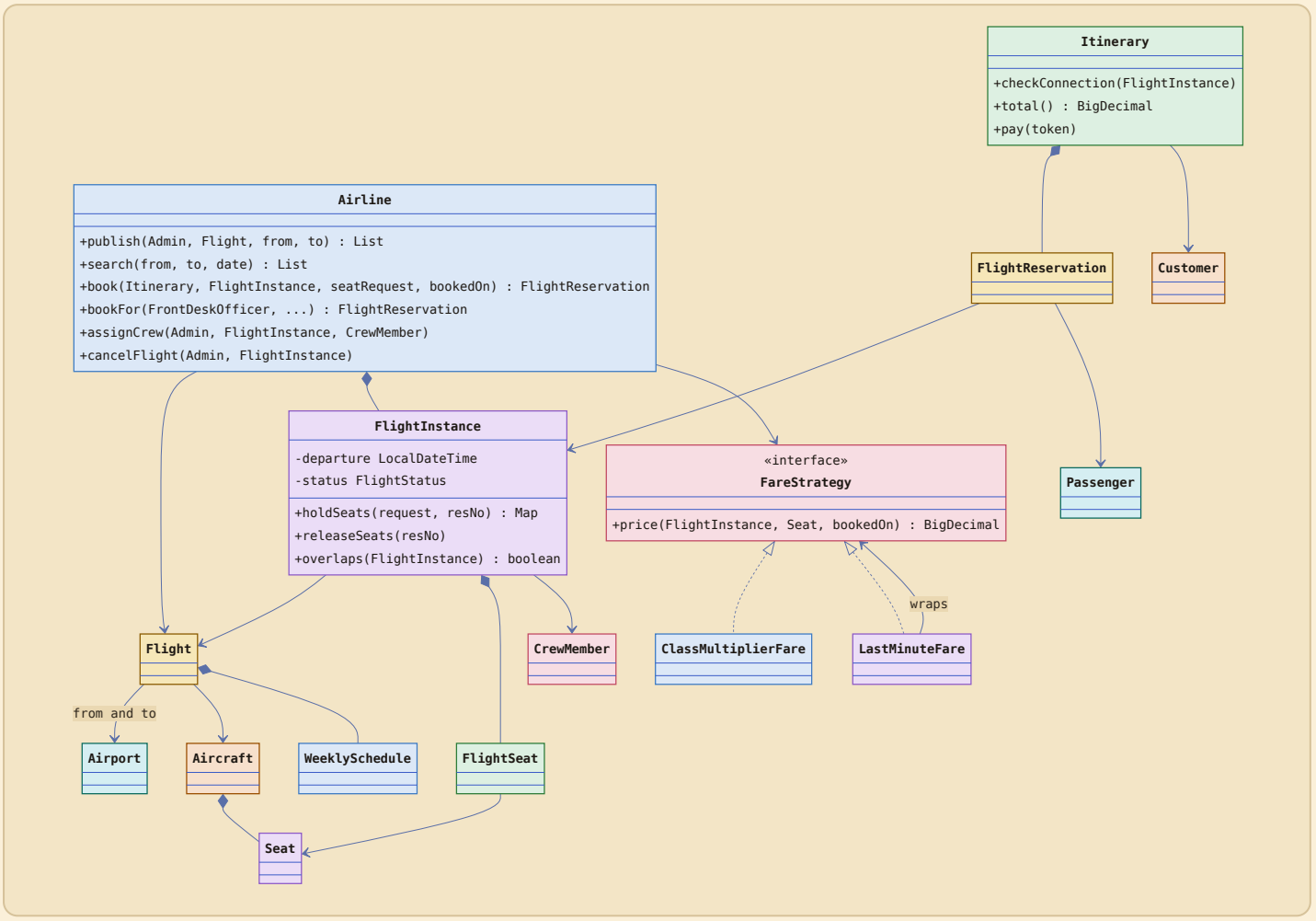


Figure — The finished class diagram of the airline management system (course diagram)

Ownership runs one way: schedules and instances belong to their flight, flight seats to their instance; airports, aircraft and crew are only referenced. `Passenger` sits outside the account tree — a passenger may never log in.



Key flows

- **Reserve:** search → choose seats and passengers → pay → confirmation. In code: `checkConnection` (same airport, ≥ 45 min after the previous leg lands) → `holdSeats` all-or-nothing → price each seat → add leg → index instance → reservation → notify.
- **Cancel reservation:** validate → cancel → release seats → notify.
- **Admin cancels a flight:** under the instance lock set `CANCELLED` and copy the index → cancel each reservation (releases seats) → notify each customer with the refund → notify every crew member.

Design decisions & patterns used

Decision	Why	Alternative rejected
Flight / FlightInstance / Seat / FlightSeat	12A is free Tuesday, taken Thursday: booking and fare belong to the seat *on an instance*.	`booked` flag on `Seat`.
`holdSeats` all-or-nothing, `synchronized` per instance	A family gets all seats or none; flights sell in parallel.	One-by-one holds; a global lock.
`Passenger` not an account	Customers book for others.	`Passenger extends Person`.
Itinerary is paid; **connection rule on `Itinerary`**	Course gap: no connecting-time check.	Pay per leg; no check.
Index instance → reservations	Course gap: cancelling notified crew but not passengers.	Scanning every itinerary.
Strategy + Decorator fares	Multipliers 1.0/1.3/1.5/2.5/4.0; `LastMinuteFare` +25% within 3 days.	Price on the seat.
Crew overlap check	No pilot on overlapping instances.	Trust the admin.
Facade `Airline`; admin ops take `Admin`	Compiler enforces "only the admin"; `bookFor` = `book` by another actor.	Rules in actors.
Course Singleton → construct and pass in	Testable.	`getInstance()`.

Java implementation

Package `lld.ch16.airline`. Full runnable code: code/lld/ch16/airline/ (see the Full Edition).

```
// FlightInstance.java — All-or-nothing: a family of three either gets all three seats or none.
synchronized Map<Passenger, Seat> holdSeats(Map<Passenger, String> request, String reservationNumber) {
    if (status != FlightStatus.SCHEDULED && status != FlightStatus.DELAYED)
        throw new IllegalStateException(key() + " is " + status);
    for (var seatNo : request.values()) {
        var fs = seats.get(seatNo);
        if (fs == null) throw new IllegalArgumentException("no seat " + seatNo + " on " + key());
        if (!fs.isFree()) return Map.of();
    }
    if (request.values().stream().distinct().count() != request.size())
        throw new IllegalArgumentException("two passengers asked for the same seat");
    Map<Passenger, Seat> out = new LinkedHashMap<>();
    request.forEach((p, no) -> { var fs = seats.get(no); fs.assign(reservationNumber); out.put(p, fs.seat()); });
    return out;
}

// Booking.java — Decorator on any fare: +25% when booked less than 3 days before departure.
public BigDecimal price(FlightInstance fi, Seat seat, LocalDate bookedOn) {
    var p = inner.price(fi, seat, bookedOn);
    long daysBefore = ChronoUnit.DAYS.between(bookedOn, fi.departure().toLocalDate());
    return daysBefore < 3 ? p.multiply(new BigDecimal("1.25")).setScale(2, RoundingMode.HALF_UP) : p;
}

// Booking.java — Itinerary: the next leg must leave from where the last one landed, with time to change.
synchronized void checkConnection(FlightInstance next) {
    if (legs.isEmpty()) return;
    var last = legs.getLast().instance();
    if (!last.flight().to().equals(next.flight().from()))
        throw new IllegalArgumentException("leg must depart from " + last.flight().to().code());
    if (next.departure().isBefore(last.arrival().plus(MIN_CONNECTION)))
        throw new IllegalArgumentException("connection too short: " + next.key() + " leaves before " + last.key()
+ " lands + 45 min");
}

// Airline.java
FlightReservation book(Itinerary it, FlightInstance fi, Map<Passenger, String> seatRequest, LocalDate bookedOn) {
    it.checkConnection(fi);
    String resNo = "PNR" + seq.incrementAndGet();
    FlightReservation r;
    synchronized (fi) {        // hold the seats AND index the reservation as one step (see cancelFlight)
        var seatMap = fi.holdSeats(seatRequest, resNo);
        if (seatMap.isEmpty()) return null;
        var amount = seatMap.values().stream().map(s -> fares.price(fi, s, bookedOn)).reduce(BigDecimal.ZERO,
BigDecimal::add);
        r = new FlightReservation(resNo, fi, seatMap, amount);
        owner.put(r, it);
        byInstance.computeIfAbsent(fi, k -> new CopyOnWriteArrayList<>()).add(r);
    }
    it.addLeg(r);
    // ... notify customer, return r
}

void cancelFlight(Admin by, FlightInstance fi) {
    List<FlightReservation> affected;
    synchronized (fi) {        // same lock as the hold+index step in book(): no booking can slip past the cancel
        fi.setStatus(FlightStatus.CANCELLED);
        affected = List.copyOf(byInstance.getOrDefault(fi, List.of()));
    }
    for (var r : affected) {
        if (r.cancel()) {
            // ... notify the itinerary's customer with the refund
        }
    }
    fi.crew().forEach(c -> notifier.send(c.id() + "@crew", "Stand down: " + fi.key() + " cancelled"));
}
```

Output excerpt:

```
   itinerary total 3300.00 (2 x 400 economy + 2 x 1250 business)
   rejected: connection too short: GA212@2026-10-06 leaves before GA100@2026-10-07 lands + 45 min
   successful bookings of 10C: 1, free seats left: 3
   economy-plus fare 500 x 1.3 = 650.00, charged 812.50
   rejected: Capt. Kim already flies GA212@2026-10-08 at that time
    [email priya@example.com] GA212@2026-10-08 is cancelled. Reservation PNR103 refunded 2500.00. Itinerary IT-101
now totals 800.00
```

20 threads race for seat 10C and exactly one wins; 650 + 25% last-minute = 812.50; the cancellation reaches every passenger through the index.

Concurrency & edge cases

- **Selling one seat twice:** `holdSeats` checks every seat, then assigns all, under the instance lock.
- **Booking while the admin cancels:** hold + index and set-`CANCELLED` + copy use the same `synchronized (fi)` — every reservation is either in the copy (cancelled and notified) or refused.
- **Double payment:** `Itinerary.pay` is idempotent. **Idempotent cancel:** `FlightReservation.cancel()` returns `false` if already cancelled — no double refund.
- **Duplicate seat in one request**, **booking a cancelled/departed flight**, **crew on overlapping instances** — all rejected.
- **Time zones:** real systems use `ZonedDateTime`/`Instant` per airport. **Holds during payment:** 10–15 min `heldUntil` on `FlightSeat` plus a sweeper.

Extensions & follow-up questions

- **Overbooking?** Per-class `overbookLimit`; assign physical seats at check-in.
- **Check-in/boarding passes, baggage, change fees, codeshares (`MarketingFlight`)** — small new entities.
- **Delays?** `DELAYED` + new departure; notify through the index; re-check connections.
- **Scale?** One row per seat with `UPDATE … WHERE reservation IS NULL` or optimistic locking; cache search.

In the interview

30-second answer: "`Flight` is the route, `FlightInstance` one departure; `Seat` is the physical chair, `FlightSeat` that chair on that departure, where booking and fare live. Seats are held all-or-nothing per instance. An itinerary groups legs, checks connections and is paid. Fares are a strategy with a last-minute decorator. An instance → reservations index lets a cancelled flight notify every passenger."

Plan: 0–5 clarify; 5–10 actors; 10–20 the four-way distinction and diagram; 20–35 code `holdSeats`, `book`, `checkConnection`, `cancelFlight`; 35–40 holds, overbooking, time zones. Expected follow-up: "How do passengers learn of a cancellation?" — "Through the instance → reservations index; without it you only reach the crew."

Self-check

1. **Q:** Where does "booked" live? **A:** On `FlightSeat` — per instance, not per aircraft.
2. **Q:** All seats or none? **A:** `holdSeats` checks all, then assigns all, in one synchronized block.
3. **Q:** `LastMinuteFare`? **A:** +25% within 3 days; a Decorator over any `FareStrategy`.
4. **Q:** Two course gaps closed? **A:** Passenger notification on cancel (index) and connection-time check (`Itinerary.checkConnection`).

TL;DR

- Flight ≠ FlightInstance; Seat ≠ FlightSeat.
- Hold seats all-or-nothing under the instance's lock.
- Itinerary = legs + connection rule + payment.
- Fares: Strategy + Decorator; instance → reservations index for cancellations.

Real-world example. Airlines separate a flight number from a leg on a date and control inventory per leg and booking class; a PNR groups passengers and segments — like `FlightReservation`/`Itinerary`. On cancellation they find every PNR on the leg, which is why the index matters.

Design a Course Registration System

Problem & clarifying questions

A **course registration system** lets students sign up for classes each term — a harsh concurrency problem: at 09:00 thousands click "register" for a few popular sections.

Terms: a **course** is a catalogue entry (CS201, 4 credits); a **section** is one offering with instructor, time and capacity (CS201-A, Mon/Wed 10:00–11:30, 30 seats) — students register for **sections** (like Flight vs FlightInstance); a **prerequisite** must be completed first; a **waitlist** is a FIFO queue for a full section; a **registration window** is when a student may register (seniors often first).

Question	Assumed answer
Courses or sections?	Sections.
Prerequisites?	Must be completed.
Capacity?	Per section; when full, a bounded waitlist.
Promotion?	Automatic on drop, FIFO, after re-checking the timetable.
Limits?	Credit limit; no time clashes; no two sections of one course.
Windows?	Open by class year; close together.
Concurrency?	Never exceed capacity; no double registration from two tabs.

Requirements

Functional: (1) register for sections and view them; (2) course has code, name, credits; section has instructor, time slot, limit; (3) search by code or name; (4) never exceed capacity, else waitlist (if room); (5) enforce prerequisites, credit limit, time conflicts, one section per course, window; (6) on drop, promote the waitlist head and notify.

Non-functional: safe concurrent registration; rules easy to add/remove ("majors only"); extensible (fees, holds, cross-listing).

Use cases

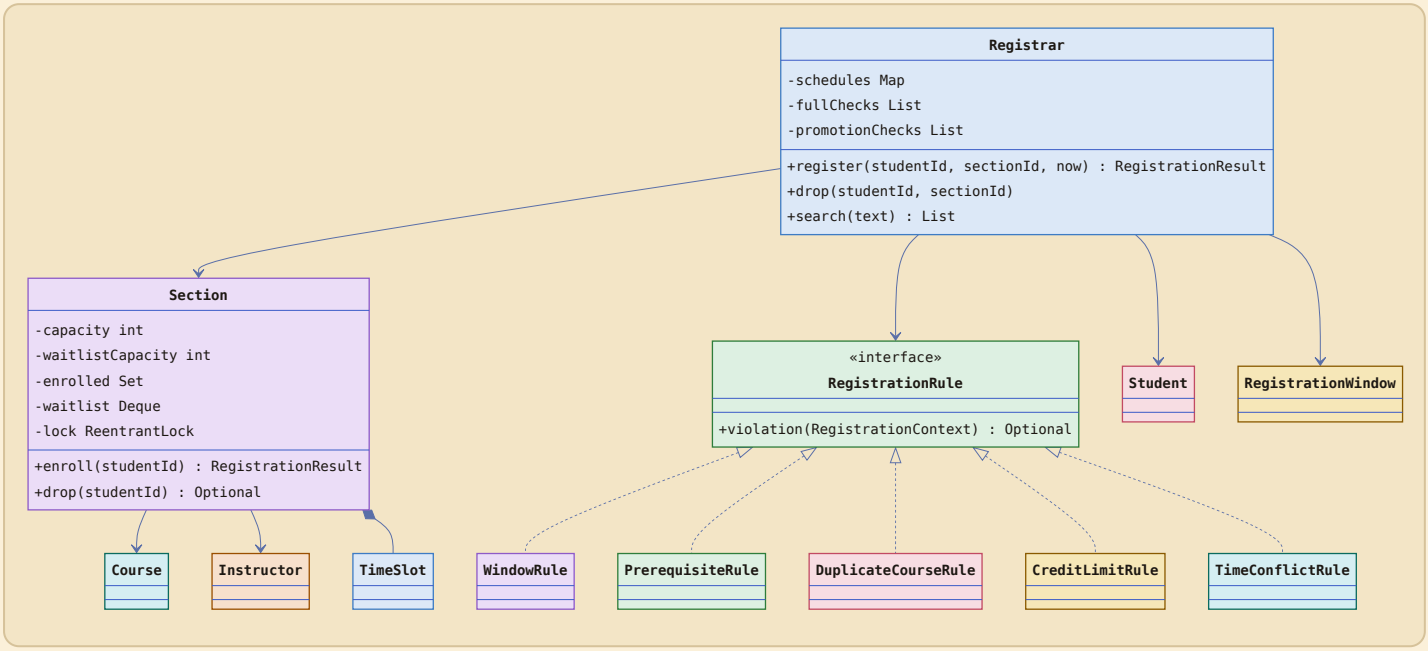
Student — search, register, drop, view schedule, join waitlist. **Registrar/Admin** — create courses/sections, set capacity and windows. **Instructor** — view roster. **System** — enforce rules, promote from waitlist, notify (a drop triggers promotion).

Core entities

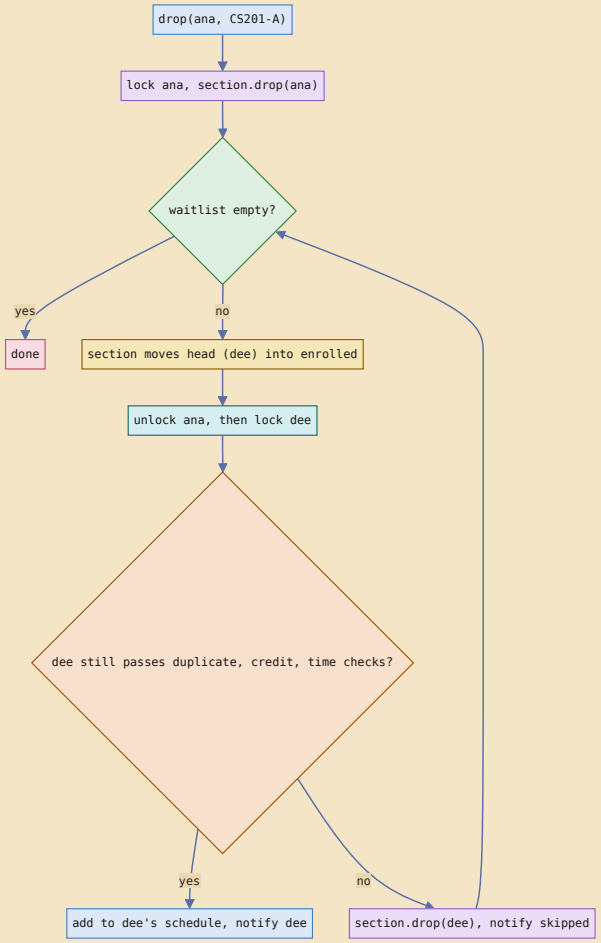
Class	Responsibility
`Course`	Code, title, credits, prerequisite codes.
`Section`	Instructor, slot, capacity, enrolled set, waitlist. **The only place a seat is taken**, under its own lock.
`TimeSlot`	Days + start/end; knows overlap.
`Instructor`, `Student`	Who teaches; year, completed courses, credit limit.
`RegistrationWindow`	Opening per class year, common close.
`RegistrationRule` (+ `WindowRule`, `PrerequisiteRule`, `DuplicateCourseRule`, `CreditLimitRule`, `TimeConflictRule`)	One rule each; returns a violation reason.
`RegistrationContext`	Everything a rule may look at.
`RegistrationResult`, `Outcome`	`ENROLLED` / `WAITLISTED` / `REJECTED` + message.
`Registrar`	Runs the rule chain, takes the seat, drops and promotes, keeps schedules, defines the lock order.

Class diagram

There is no course lesson for this problem; it comes from the *awesome-low-level-design* problem set (register, view, search, capacity, concurrency, extensibility) — all covered, with instructor and capacity on `Section`.



Key flows



- **Register:** lock the student's schedule → run window, prerequisites, duplicate, credits, time conflict → first failure = `REJECTED` + reason; else `section.enroll` (section lock) → `ENROLLED` (add to schedule), `WAITLISTED at position n`, or `REJECTED` (both full).
- **Student status in a section:** `ENROLLED`, `WAITLISTED` (→ `ENROLLED` on promotion, or `SKIPPED` if the re-check fails), `REJECTED`; drop leaves.
- **Drop and promote:** as in the flowchart above.

Design decisions & patterns used

Decision	Why	Alternative rejected
Course vs Section	Capacity, time, instructor belong to an offering.	Capacity on `Course`.
Seat taken only in `Section.enroll`, `ReentrantLock` per section	Atomic check-capacity-then-add; sections don't block each other.	`synchronized` registrar (the whole university in one line).
Per-student lock around rules + enroll	Credit/time checks read the schedule; two tabs would both pass without it.	Section lock only.
Lock order: student, then section	Deadlock needs opposite orders; promotion takes the next student's lock only *after* releasing the dropper's.	Promote while holding the dropper's lock.
Rules as objects (Specification / Chain of Responsibility)	A new rule = new class + list entry (Open/Closed); promotion reuses a subset.	One long `if` method.
Bounded FIFO `Deque` waitlist	Fair, bounded.	Unbounded list.
Re-check on promotion	Timetable may have changed while waiting.	Promote blindly.
`now` passed in	Testable windows.	`LocalDateTime.now()` in rules.
GitHub problem's **Singleton** → one `Registrar` passed in	Testable.	`getInstance()`.

Java implementation

Package `lld.ch16.courseregistration`. Full runnable code: code/lld/ch16/courseregistration/ (see the Full Edition).

```
// Section.java — The only place a seat is taken: check-and-add in one critical section.
RegistrationResult enroll(String studentId) {
    lock.lock();
    try {
        if (enrolled.contains(studentId)) return RegistrationResult.rejected("already enrolled in " + id);
        if (waitlist.contains(studentId)) return RegistrationResult.rejected("already on the waitlist of " + id);
        if (enrolled.size() < capacity) {
            enrolled.add(studentId);
            return new RegistrationResult(Outcome.ENROLLED, "enrolled in " + id + " (" + enrolled.size() + "/" +
capacity + ")");
        }
        if (waitlist.size() < waitlistCapacity) {
            waitlist.addLast(studentId);
            return new RegistrationResult(Outcome.WAITLISTED, "waitlisted for " + id + " at position " +
waitlist.size());
        }
        return RegistrationResult.rejected(id + " and its waitlist are full");
    } finally {
        lock.unlock();
    }
}

// Rules.java
interface RegistrationRule { Optional<String> violation(RegistrationContext c); }

final class CreditLimitRule implements RegistrationRule {
    public Optional<String> violation(RegistrationContext c) {
        int current = c.currentSchedule().stream().mapToInt(s -> s.course().credits()).sum();
        int after = current + c.section().course().credits();
        return after <= c.student().maxCredits() ? Optional.empty()
                : Optional.of("credit limit " + c.student().maxCredits() + " exceeded (" + after + ")");
    }
}

// Registrar.java — lock order is ALWAYS student-then-section
RegistrationResult register(String studentId, String sectionId, LocalDateTime now) {
    // ... look up student, section (reject if unknown)
    var schedule = schedules.get(studentId);
    synchronized (schedule) {                                   // 1st lock: the student
        var ctx = new RegistrationContext(student, section, Set.copyOf(schedule), now, window);
        var v = firstViolation(fullChecks, ctx);
        if (v.isPresent()) return RegistrationResult.rejected(v.get());
        var result = section.enroll(studentId);                 // 2nd lock: the section
        if (result.outcome() == Outcome.ENROLLED) schedule.add(section);
        return result;
    }
}

void drop(String studentId, String sectionId) {
    // ...
    synchronized (schedule) {
        promoted = section.drop(studentId);
        schedule.remove(section);
    }
    notifier.accept(studentId + " dropped " + sectionId);
    // Finish promotions OUTSIDE the dropper's lock, one student lock at a time (no nested student locks).
    while (promoted.isPresent()) promoted = finishPromotion(promoted.get(), section);
}
```

`finishPromotion` locks the promoted student, re-runs `promotionChecks` (duplicate, credit, time), and either adds the section to their schedule or drops them and returns the next promotion.

Output excerpt:

```
   ana  MATH201-A         -> REJECTED: time conflict with CS201-A [MONDAY, WEDNESDAY] 10:00-11:30
   ana  CS301-A           -> REJECTED: missing prerequisites [CS201, MATH201]
   fay  CS201-A           -> REJECTED: CS201-A and its waitlist are full
   ben  MATH201-A         -> REJECTED: credit limit 4 exceeded (7)
    [notify] dee promoted from waitlist into CS201-A
5) 50 first-years grab 30 seats (+10 waitlist) in CS101-A at 09:00 sharp
   ENROLLED=30 WAITLISTED=10 REJECTED=10, roster size 30
6) One student double-clicks: two threads, same request
   {ENROLLED=1, REJECTED=1}, gus schedule [CS201-B]
```

The 09:00 rush counts are exact every run; the double-click succeeds exactly once.

Concurrency & edge cases

- **Overselling** — section lock around "size < capacity → add". **Double-click / two tabs** — per-student lock; the second request hits `DuplicateCourseRule` or "already enrolled".
- **Deadlock** — fixed order student → section; never two student locks at once.
- **Waitlisted in one section, enrolled in another of the same course** — caught by the promotion re-check, which skips to the next student.
- **Window closes while waitlisted** — promotion doesn't re-check the window; add `WindowRule` to `promotionChecks` if that is the policy.
- Dropping a section you're not in → `IllegalArgumentException`; windows are half-open (`opens ≤ now < closes`).
- **Beyond one JVM:** `UPDATE section SET enrolled = enrolled + 1 WHERE id = ? AND enrolled < capacity` (atomic conditional update) or optimistic locking; a queue in front of hot sections.

Extensions & follow-up questions

- **Co-requisites / "prereq OR permission"?** Composite expression tree (`And`, `Or`, `CourseCompleted`, `Permission`) in `PrerequisiteRule`.
- **Holds?** A `HoldRule`. **Priority waitlist?** `PriorityQueue` by (year, time joined). **Major seat buckets?** Split capacity in `Section.enroll`.
- **Lecture + lab?** Lock both sections by id order, check both, enrol both. **Notifications/fees?** Swap the `Consumer<String>` notifier; a `Bill` per student per term.

In the interview

30-second answer: "Students register for sections, not courses. A section owns capacity and a bounded FIFO waitlist and takes seats only inside its own lock. The registrar runs a chain of rule objects under the student's lock, then asks the section for a seat. Lock order is always student then section, so no deadlocks. On drop the waitlist head is promoted after a re-check."

Plan: 0–5 clarify; 5–12 entities and diagram; 12–30 code `Section.enroll/drop`, the rule interface and two rules, `register`; 30–40 concurrency, lock order, DB conditional update.

Self-check

1. **Q:** Which two locks does `register` take, in what order? **A:** Student's schedule, then the section — always.
2. **Q:** Why not `synchronized` on the whole registrar? **A:** Correct but serialises the whole university.
3. **Q:** Why re-check on promotion? **A:** The timetable may have changed while waiting.
4. **Q:** Capacity with many servers? **A:** Atomic conditional update (`… WHERE enrolled < capacity`) or optimistic locking.

TL;DR

- Course (catalogue) vs Section (capacity, slot, waitlist).
- Rules are small objects run as a chain.
- Seats taken only in `Section.enroll`; student lock first, section lock second.

- Drop → promote head → re-check → notify. At scale: DB conditional update.

Real-world example. Student-information systems such as Ellucian Banner run priority windows by class standing, check prerequisites and clashes, and keep section waitlists; registration-day slowdowns are the 09:00 "thundering herd" this design handles.

Chapter summary

Problem	Key insight	Atomic claim	Patterns
Car rental	Availability is about **dates**, not status	`Vehicle.tryBook(range)`	Template Method, Strategy + Decorator, state machine
Restaurant	**Meal per seat**; table = status + slots	`Table.tryHold(time)`	Observer, Strategy (chefs, payment), state machine
Airline	**Flight ≠ FlightInstance, Seat ≠ FlightSeat**	`FlightInstance.holdSeats(map)` (all or nothing)	Strategy + Decorator, Facade, cancellation index
Course registration	**Course ≠ Section**; rules as objects	`Section.enroll` inside a student lock	Specification / Chain of rules, lock ordering

Find the calendar, find the atomic check-then-claim, and put the rules on the object that owns the data.